# MATHEMATICS

***Maximum Marks: 80***

***Time allowed: Three hours***

1. *Answers to this Paper must be written on the paper provided separately.*

2. *You will **not** be allowed to write during first **15** minutes.*

3. *This time is to be spent in reading the question paper.*

4. ***The time given at the head of this Paper is the time allowed for writing the answers.***

5. *Attempt **all** questions from **Section A** and **any four** questions from **Section B.***

6. ***All working, including rough work, must be clearly shown, and must be done on the same sheet as the rest of the answer.***

7. ***Omission of essential working will result in loss of marks.***

8. *The intended marks for questions or parts of questions are given in brackets [ ]*

9. ***Mathematical tables and graph papers are to be provided by the school.***

***Instruction for the Supervising Examiner***

*Kindly read aloud the Instructions given above to all the candidates present in the Examination Hall.*

**This paper consists of 15 printed pages and 1 blank page.**

## SECTION A (40 Marks)

*(Attempt **all** questions from this **Section**.)*

**Question 1**

Choose the correct answers to the questions from the given options. [15]

(Do not copy the questions, write the correct answers only.)

(i) The marked price of a smart TV is ₹ 18,000. The rate of GST is 18%. The price paid by the customer for the smart TV is:

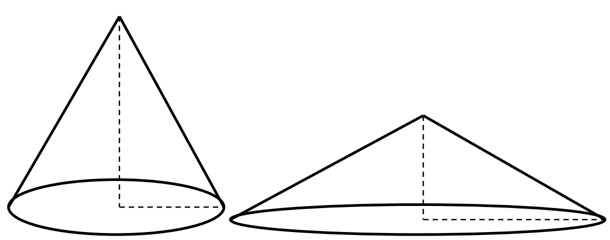
(a) ₹12,960

(b) ₹19,620

(c) ₹18,000

(d) ₹21,240

(ii) Two right circular cones, $C_1$ and $C_2$, have their heights in the ratio 2 : 1 and radii in the ratio 1 : 2. The ratio of their volumes is:



$C_1$ $C_2$

(a) 1 : 4

(b) 1 : 2

(c) 2 : 1

(d) 4 : 1

(iii) A person deposited ₹500 per month in a recurring deposit account for one year at *r*% per annum. The rate of interest is revised to *2r*%. The interest earned by the person compared to the original amount of interest will be:

(a) one-fourth

(b) unchanged

(c) half

(d) double

(iv) A company declares 9% dividend on its shares of ₹100 available at ₹120. It means a person will receive annual dividend which is 9% of:

(a) market value

(b) face value

(c) the difference of market value and face value

(d) company's profit

(v) The discriminant of the quadratic equation $2x^2 + 5x + c = 0$ is 1. The value of $c$ is:

(a) -3

(b) 0

(c) 2

(d) 3

(vi) $(x + 3)$, $(x + 6)$ and $(x + 10)$ are in continued proportion. The value of $x$ is:

(a) 3

(b) 6

(c) 9

(d) 12

(vii) $(x - 1)$ is a factors of the polynomial $x^3 + 2x^2 - x - k$. The value of $k$ is:

(a) $-2$

(b) $-1$

(c) 1

(d) 2

(viii) Matrix $A = \begin{bmatrix} -1 & 2 \\ 3 & 0 \end{bmatrix}$ and the Matrix $mA = \begin{bmatrix} 3 & -6 \\ -9 & 0 \end{bmatrix}$. The value of $m$ is:

(a) $-\frac{1}{3}$

(b) $-3$

(c) $\frac{1}{3}$

(d) 3

(ix) The points $A\ (x, 0), B\ (-1, -1)$ and $C\ (0, y)$ are collinear. The value of $\frac{1}{x} + \frac{1}{y}$ is:

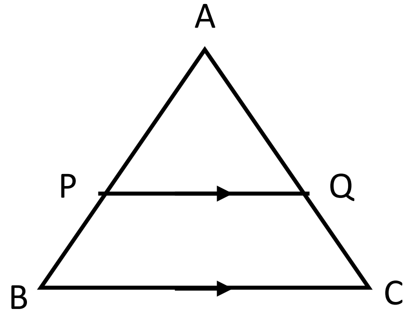
(a) $-1$

(b) 0

(c) 1

(d) $xy$

(x) In the given diagram, $PQ\ //BC$ and $ar\Delta APQ = ar\ quad\ BPQC$.




The value of *AP : PB* is:

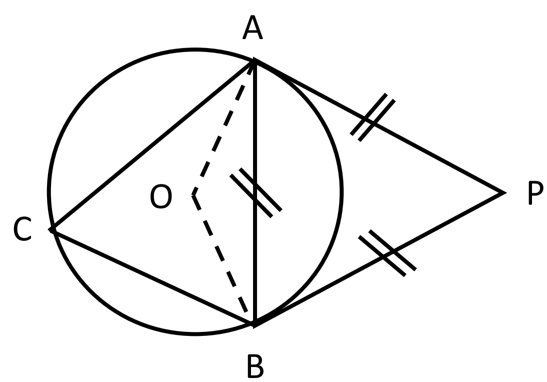
(a) 1 : 2

(b) 2 : 1

(c) $1 : \sqrt{2}$

(d) $(\sqrt{2} + 1)\ : 1$

(xi) In the given diagram, PA and PB are the tangents to the circle with centre O, such that PA = PB = AB. The value of ∠ACB is:



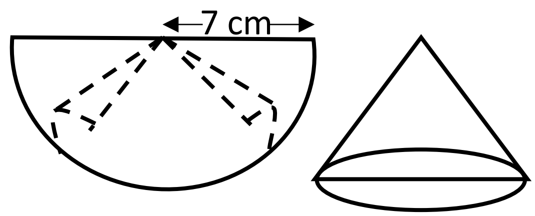

(a) 30°

(b) 45°

(c) 60°

(d) 90°

(xii) A semi - circular sheet of paper having radius 7 cm is folded to form a right circular cone, as shown in the diagram. The slant height of the cone so formed is:




(a) 3.5 cm

(b) 7 cm

(c) $7\sqrt{2}$ cm

(d) 14 cm

(xiii) **Assertion (A) :** $sec\theta(1 + sin\theta)(sec\theta + tan\theta) = 1$

**Reason (R) :** $sec^2\theta - tan^2\theta = 1$, for any value of $\theta$.

(a) (A) is true, (R) is false.

(b) (A) is false, (R) is true.

(c) Both (A) and (R) are true and (R) is the correct explanation of (A).

(d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(xiv) **Assertion (A) :** The probability of Sun rising from the west is 0.

**Reason (R) :** The probability of an impossible event is always 0.

(a) (A) is true, (R) is false.

(b) (A) is false, (R) is true.

(c) Both (A) and (R) are true and (R) is the correct explanation of (A).

(d) Both (A) and (R) are true but, (R) is not the correct explanation of (A).

(xv) The mean of $x_1, x_2, x_3, \ldots\ldots\ldots, x_n$ is $m$. The value of each variate is increased by 2. The mean of the new data is:

(a) $m - 2$

(b) $m$

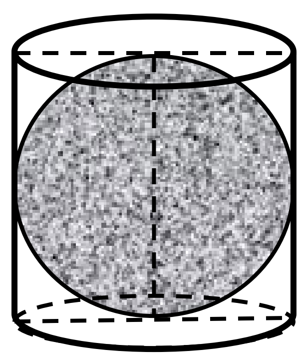
(c) $m + 2$

(d) $2m$

**Question 2**

(i) A solid sphere completely fits into a cylindrical container of radius 3.5 cm and height 7 cm as shown in the figure below. Find the volume of water required to fill the remaining (unshaded) space of the container to the nearest of a whole number. [4]

$\left(use\ \pi = \frac{22}{7}\right)$



(ii) A car covers a distance of 400 km at a certain speed. Had the speed been 12 km/h more, the time taken for the journey would have been 1 hour 40 minutes less. Find the: [4]

(a) original speed of the car.

(b) time taken with the increased speed.

(iii) The coordinates of the vertex of $\Delta ABC$ are $A\ (a, b), B\ (5, -3)$ and $C(-5,3)$. The coordinates of the centroid $G$ of $\Delta ABC$ is $G\ \left(\frac{2}{3}, \frac{1}{3}\right)$. Find the: [4]

(a) coordinates of the vertex *A*.

(b) equation of median through the vertex *C*.

**Question 3**

(i) Anamika deposited ₹300 per month in a cumulative deposit account with a bank for 3 years. Her brother Sanjeev started depositing ₹500 per month for 2 years in the same scheme. The banks paid 10% simple interest per annum to both. At the time of maturity, find: [4]

(a) the interest earned by Anamika.

(b) who got more money as interest and how much?

(ii) $\sqrt{2}, \sqrt{8}, \sqrt{18}$ ... ... forms a progression. [4]

(a) Identify the type of progression.

(b) The sum of its first 10 terms is $p\sqrt{2}$. Find the value of *p*.

(iii) Use graph sheet for this question. Take 2 cm = I unit along the axes. Plot and write coordinates of : [5]

(a) A (2, 3) and B (4, 5).

(b) image of A and B in the *x*-axis as A′ and B′.

(c) image of A′ in the *y*-axis as A″.

(d) name the single transformation that maps A to A″.

## SECTION B (40 Marks)

*(Attempt **any four** questions from this Section.)*

**Question 4**

(i) A computer mechanic charges repairing cost of different components as tabulated below: [3]

| Component | Repairing cost (₹) | Discount (%) |
|---|---|---|
| A | ₹5000 | 20% |
| B | ₹6500 | 30% |

The rate of GST is 18%. Find the total:

(a) discount

(b) amount of GST

(ii) Given, matrix A = $\begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$ and I is a unit matrix of order 2 x 2. [3]

(a) Find the matrix, $A^2$ *and* $5A$

(b) value of $k$, such that $A^2 - 5A = k.I$

(iii) Prove that: [4]

$$\sqrt{\frac{sec\theta + 1}{sec\theta - 1}} + \sqrt{\frac{sec\theta - 1}{sec\theta + 1}} = 2\ cosec\theta$$

**Question 5**

(i) Prasoon invests ₹7500 in a company paying 6% dividend annually. ₹100 shares of this company are available at ₹150. [3]

(a) Find the number of shares he purchased.

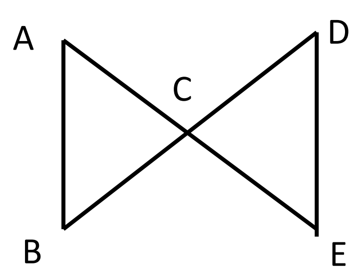
(b) Find his annual income.

(c) After one year, he sold 50% of his original shares at ₹175 each. Find his gain on the shares he sold.

(ii) Solve the following inequation, write the solution set and represent it on the real number line. [3]

$$3x + \frac{14}{3} > \frac{4x}{3} - 2 \geq 3x - 4\,,\, x \in R$$

(iii) In the given diagram, lines AE and BD intersect each other at C such that ΔABC ~ ΔDEC. [4]




(a) Complete the equation with corresponding sides of ΔABC and ΔDEC.

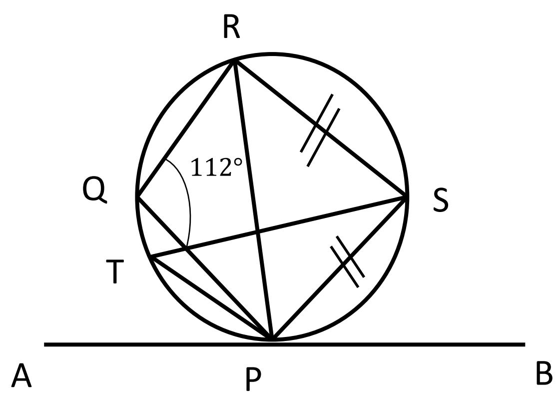
$$\frac{AB}{DE} = \frac{AC}{?} =? \frac{?}{EC}$$

(b) ∠ABC is equal to which angle of ΔDEC?

(c) If, AB : DE = 2 : 3, find the ratio of *ar (ΔABC)* : *ar (ΔDEC).*

**Question 6**

(i) In the given diagram, AB is a tangent to the circle at P. If $\angle PQR = 112°$ and PS = SR, find: [3]




(a) $\angle PSR$

(b) $\angle BPR$

(c) $\angle PTS$

(ii) Using step – deviation method, find mean for the following data. [3]

Give your answer to the nearest of a whole number.

| **Marks** | 20 – 25 | 25 – 30 | 30 – 35 | 35 – 40 | 40 – 45 | 45 - 50 |
|---|---|---|---|---|---|---|
| **No. of Students** | 10 | 7 | 9 | 13 | 5 | 6 |

(iii) The remainder obtained by dividing the polynomial $m\,x^2 - 3x + 6$ by $(x - 2)$ is twice the remainder obtained by dividing the polynomial $3\,x^2 + 5x - m$ by $(x + 3)$. [4]

(a) Find the value of *'m'*.

(b) Using the value of *'m'*, find the remainder when $mx^2 - 3x + 6$ is divided by $(x - 2)$.

**Question 7**

(i) Each of the letters of the word 'BINOCULARS' is written on identical cards and put in a bag. They are well shuffled. If a card is drawn at random from the bag, find the probability that the letter is: [3]

(a) a vowel

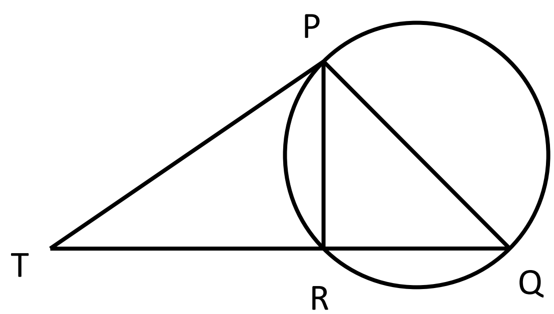
(b) one of the first 8 letters of the English alphabet which appears in the given word.

*(Support your working with at least one sample space)*

(ii) Using properties of proportion, solve for $x$: [3]

$$\frac{\sqrt{2x+3}+\sqrt{2x-3}}{\sqrt{2x+3}-\sqrt{2x-3}}=3$$

(iii) In the given diagram, PT is a tangent to the circle and QT is a secant meeting the circle at R, such that QR = 9 cm and RT = 16 cm. [4]




(a) Prove that: $\Delta PTR \sim \Delta QTP$.

(b) Find the length of PT.

**Question 8**

(i) Use graph paper for this question; draw the histogram and estimate mode for the following data. Use 2 cm = 10 units along the *x*-axis, and 2 cm = 2 units along the *y*-axis. [3]

| Class | 30 - 40 | 40 - 50 | 50 – 60 | 60 – 70 | 70 – 80 |
|---|---|---|---|---|---|
| Frequency | 10 | 12 | 18 | 9 | 11 |

(ii) Solve the quadratic equation $x^2 - 6x - 8 = 0$ and give your answer correct to three significant figures. [3]

*(Use mathematical tables for this question, if necessary)*

(iii) The surface area of a solid metallic sphere is $616\ cm^2$. It is melted and recast into solid right circular cones of diameter 3.5 cm and height 3.5 cm. Find the: [4]

(a) radius of the sphere.

(b) number of cones formed.

$\left(use\ \pi = \frac{22}{7}\right)$

**Question 9**

(i) As observed from the top of a light house, 80 m above sea level, the angle of depression of a ship, sailing directly towards it, changes from 35° to 55°. [5]

(a) Draw a free hand sketch representing the given information.

(b) Find the distance travelled by the ship during the period of observation.

Give your answer to the nearest meter.

(Use mathematical tables for this question)

(ii) Use graph sheet for this question. Take 2 $cm$ = 10 students along one axis and 2 $cm$ = 5 $cm$ of height along the other axis. [5]

The following table shows the height of 80 students in a class.

| Height (in cms) | 150 – 155 | 155 – 160 | 160 – 165 | 165 – 170 | 170 – 175 | 175 – 180 |
|---|---|---|---|---|---|---|
| No. of students | 8 | 15 | 24 | 17 | 11 | 5 |

Draw the cumulative frequency curve for the above distribution and find the:

(a) median height.

(b) lower – quartile height.

(c) number of students whose height is more than 173 cm.

**Question 10**

(i) The sum of first three terms of a Geometric Progression (G.P.) is $\frac{21}{2}$ and their product is 27. Given $r < 1$, find its: [3]

(a) common ratio.

(b) first three terms.

(ii) The equation of a line is $3x - 4y + 7 = 0$. Find the: [3]

(a) slope of the line.

(b) equation of a line perpendicular to the given line and having a $y$-intercept equals to 2.

(iii) Using a ruler and compass only, construct: [4]

(a) $\Delta ABC$ such that AB = 5 cm, AC = 6.3 cm and $\angle ABC = 60°$.

(b) the circumcircle of $\Delta ABC$.

(c) the locus of points which are equidistant from AB and BC.